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Credit Card Fraud Detection Using ML

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MODULE: Artificial Intelligence(EIARI4A)

DATE: 18 May 2026

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CREDIT CARD FRAUD

- Annual global losses: > $30 BILLION
- Affects banks, merchants, and consumers
- Fraud patterns evolve continuously

LIMITATIONS OF TRADITIONAL SYSTEMS

- Rule-based systems cannot adapt to new fraud patterns
- High false positive rate
- High maintenance cost

RESEARCH QUESTION

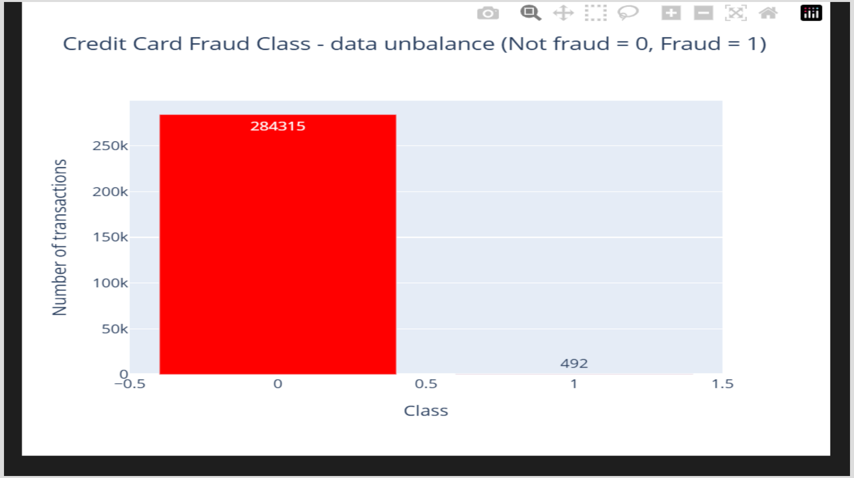
Can machine learning detect fraud more accurately than traditional rule-based systems?

Source: Kaggle Credit Card Fraud Detection Dataset
Period: September 2013 (2 consecutive days)

Total Transactions: 284,807
Fraudulent Transactions: 492 (0.172%)
Non-Fraud Transactions: 284,315 (99.828%)

FEATURES:
- V1 - V28: PCA-transformed
- Time
- Amount
- Class

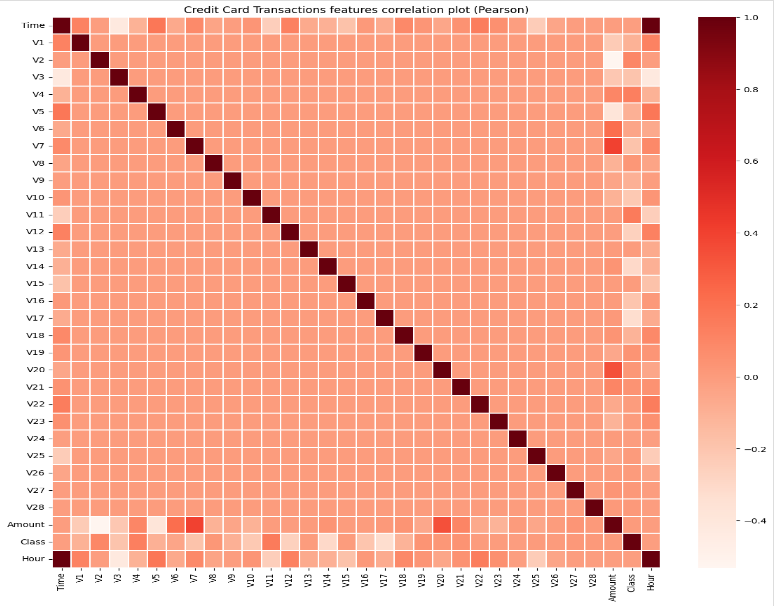
- Non-Fraud: 99.83%
- Fraud: 0.17%
- A naive model predicting non-fraud for all transactions would appear highly accurate but detect zero fraud.



Original confidential transaction features were transformed using PCA.

V1-V28:

- Ordered by variance
- Centered around zero
- Uncorrelated
- Exact business meaning unknown



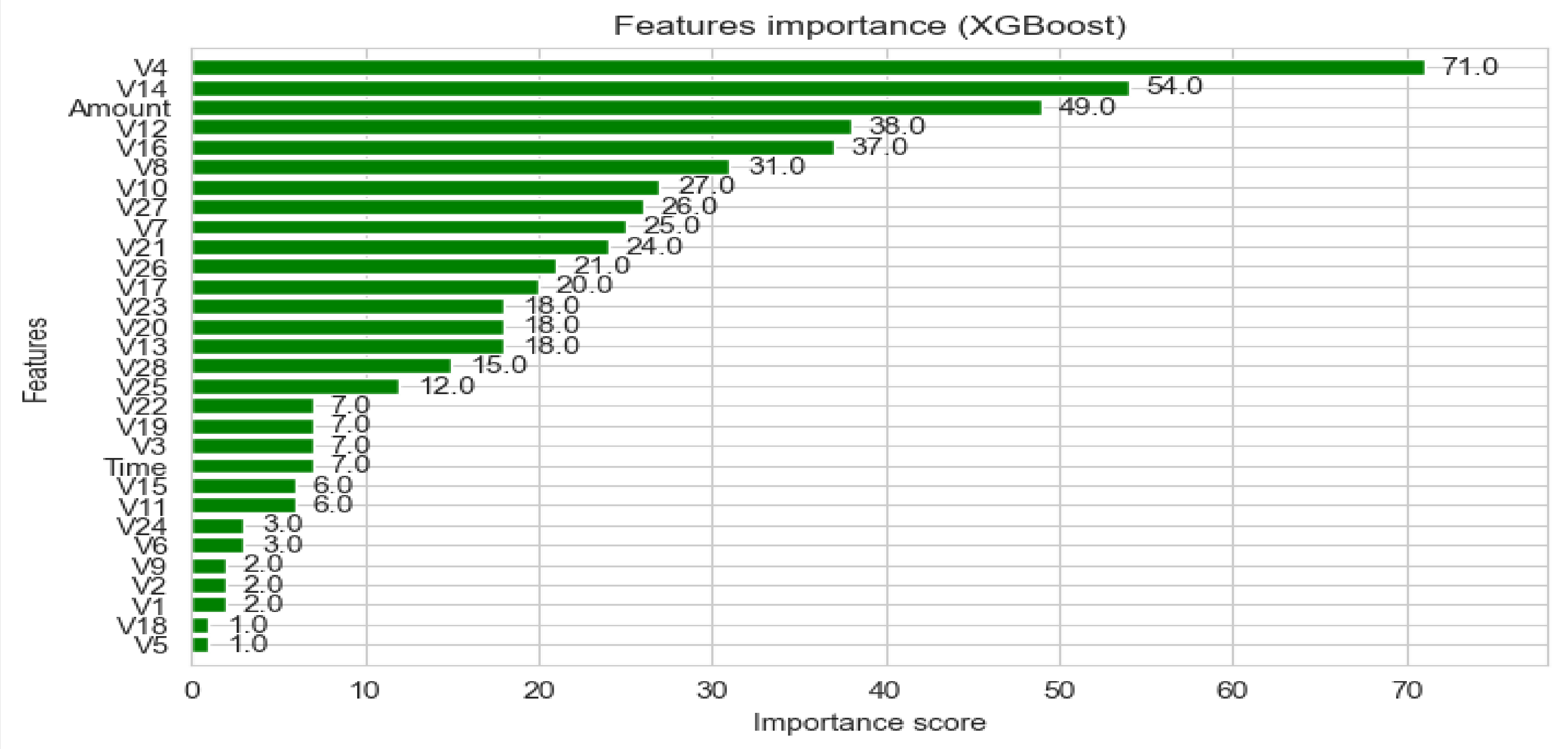
DATA PREPROCESSING:

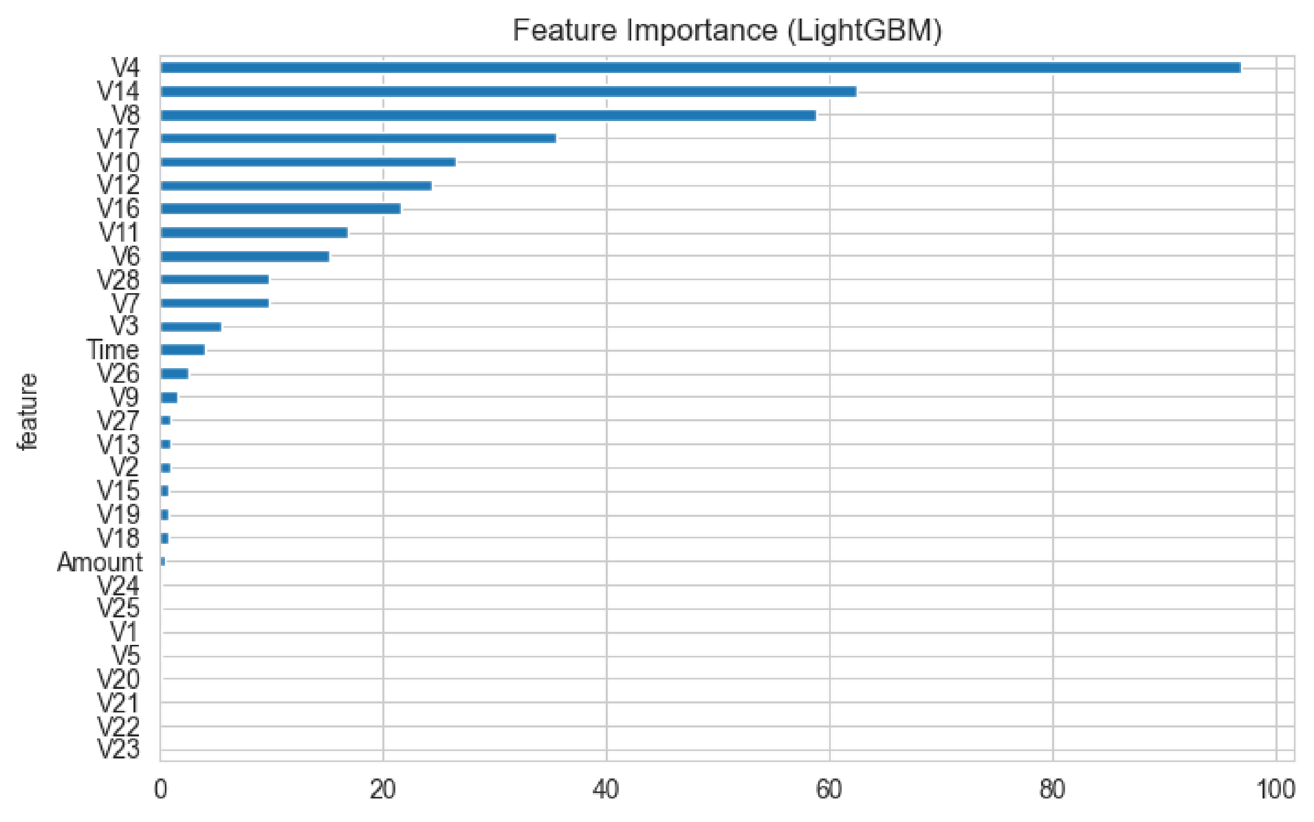
- 70/15/15 train-validation-test split
- StandardScaler on Amount
- Dropped Time feature
- No SMOTE used

MODELS:

- Logistic Regression
- Random Forest
- AdaBoost
- CatBoost
- XGBoost
- LightGBM

- Logistic Regression - Baseline model
- Random Forest - Bagging ensemble
- AdaBoost - Sequential boosting
- CatBoost - Efficient categorical handling
- XGBoost - Best performer
- LightGBM - Fast boosting framework


Features importance (XGBoost)
V4
V14
Amount
V12
V16
V8
V10
V27
V7
V21
V26
V17
V23
V20
V13
V28
V25
V22
V19
V3
Time
V15
V11
V24
V6
V9
V2
V1
V18
V5
71.0
54.0
49.0
38.0
37.0
31.0
27.0
26.0
25.0
24.0
21.0
20.0
18.0
18.0
18.0
15.0
12.0
7.0
7.0
7.0
7.0
6.0
6.0
3.0
3.0
2.0
2.0
2.0
1.0
1.0
0
10
20
30
40
50
60
70
Importance score
Features


Feature Importance (LightGBM)
feature
V4
V14
V8
V17
V10
V12
V16
V11
V6
V28
V7
V3
Time
V26
V9
V27
V13
V2
V15
V19
V18
Amount
V24
V25
V1
V5
V20
V21
V22
V23
0
20
40
60
80
100

VUT

Results - Model Comparison

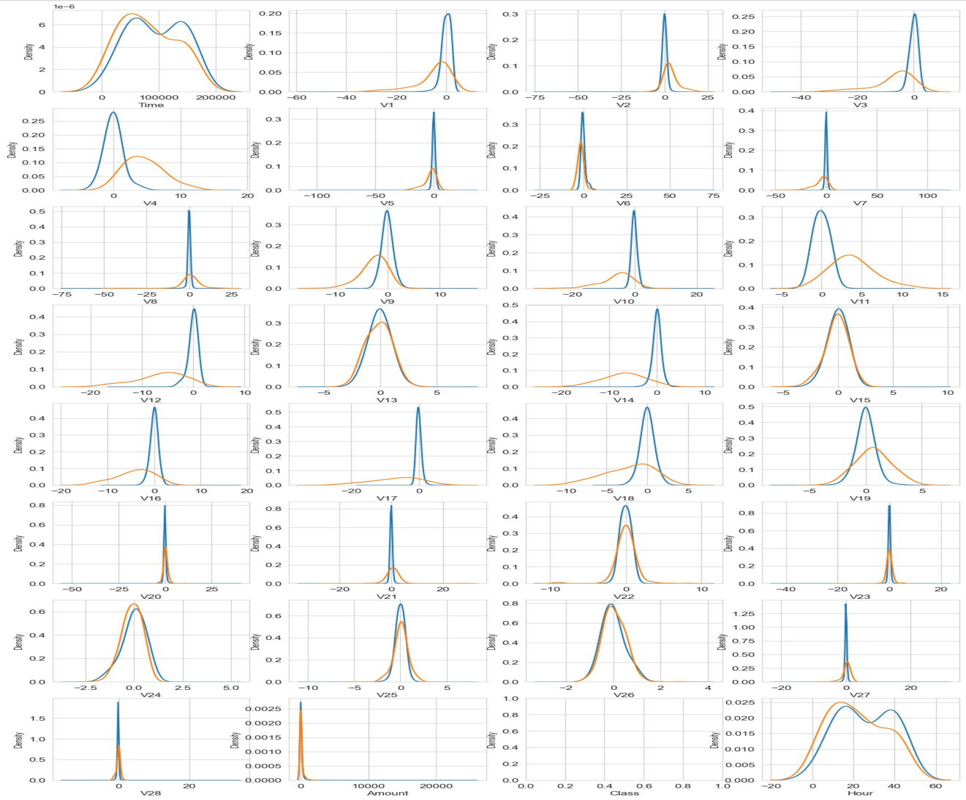
- XGBoost: Test AUC 0.9778
- LightGBM: Test AUC 0.9469
- CatBoost: Test AUC 0.8578
- Random Forest: Test AUC 0.8529
- AdaBoost: Test AUC 0.8332

- XGBoost selected as best model.

XGBoost Training Progress

- Best Validation AUC: 0.9845
- Best round: 332
- Early stopping prevented overfitting
- Inference time: <50ms

Random Forest Trees



LightGBM Cross-Validation

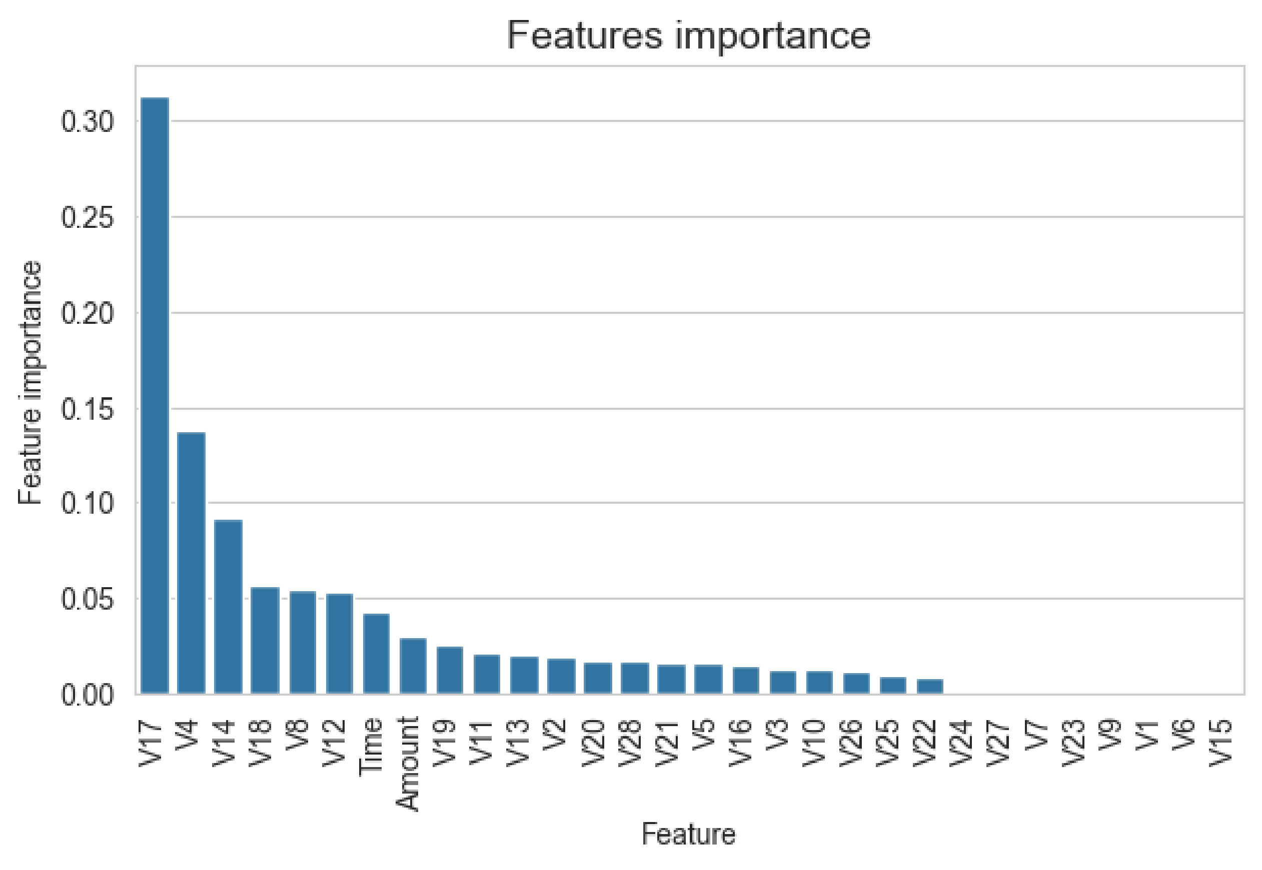
- 5-Fold Cross Validation Results:
- Fold 1: 0.9946
- Fold 2: 0.9696
- Fold 3: 0.9487
- Fold 4: 0.9871
- Fold 5: 0.9933

- Mean CV AUC: 0.9431

Feature Importance

- Most important features:
- •V14
- •V10
- •V12
- •V4
- •V17

- Time and Amount had low importance.


Features importance
Feature importance
0.30
0.25
0.20
0.15
0.10
0.05
0.00
V17
V4
V14
V18
V8
V12
Time
Amount
V19
V11
V13
V2
V20
V28
V21
V5
V16
V3
V10
V26
V25
V22
V24
V27
V7
V23
V9
V1
V6
V15
Feature

Financial Impact Analysis

- Estimated annual savings: $9.4 million
- Fraud recall: 93%
- False positive rate: 0.09%

- Net monthly savings: $786,000

Deployment Strategy

- Phase 1: Shadow deployment
- Phase 2: Flag and review
- Phase 3: Auto-decline high-risk transactions
- Phase 4: Continuous maintenance and retraining

Key Findings

- • XGBoost achieved highest performance
- • Tree-based models handled imbalance effectively
- • Real-time fraud detection is feasible
- • Annual savings potential is significant

Limitations

- • PCA-transformed features reduce interpretability
- • Dataset from 2013
- • Limited temporal scope
- • Moderate precision with false alarms

Future Work

- • Hyperparameter optimization
- • Ensemble methods
- • SHAP interpretability
- • Real-time streaming
- • Concept drift detection

Conclusion

- XGBoost achieved 0.9778 test AUC.
- The system is production-ready for real-time fraud detection.
- Top fraud indicators: V14, V10, V12.
- Projected annual savings: $9.4 million.

References

- [1] Breiman, L. (2001). Random forests.
- [2] Chen, T., & Guestrin, C. (2016). XGBoost.
- [3] Chawla, N.V., et al. (2002). SMOTE.
- [4] Dal Pozzolo, A., et al. (2015). Credit card fraud detection.

Thank You & Q&A

- THANK YOU

- Questions & Discussion

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Thank you

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